

# Modelos generativos

GANs y VAEs

# GAN: Generative Adversarial Nets



*“Adversarial training is the coolest idea in Machine Learning in the last 20 years”*

**Yann Lecun**

Vice President, Chief AI Scientist en Facebook

Professor at New York University

Founding director of the NYU Center for Data Science

Turing Award 2018

# Real or Fake?

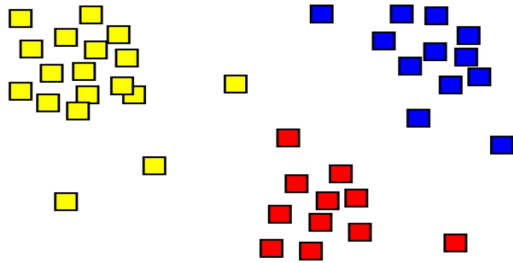


# Real or Fake?



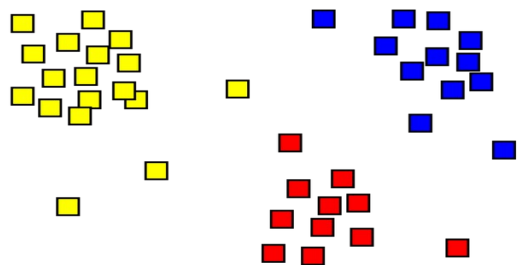
# Aprendizaje no supervisado

Clustering: encontrar grupos en los datos (colores)



# Aprendizaje no supervisado

Clustering: encontrar grupos en los datos (colores)



Modelos generativos: encontrar la función  $P(X)$  que generó los datos

Dos tipos: Explícitos e Implícitos

*“What I cannot create, I do not understand.”*

—Richard Feynman

Obtener  $p(x)$

Samplear  $p(x)$

# 2014



---

## Generative Adversarial Nets

---

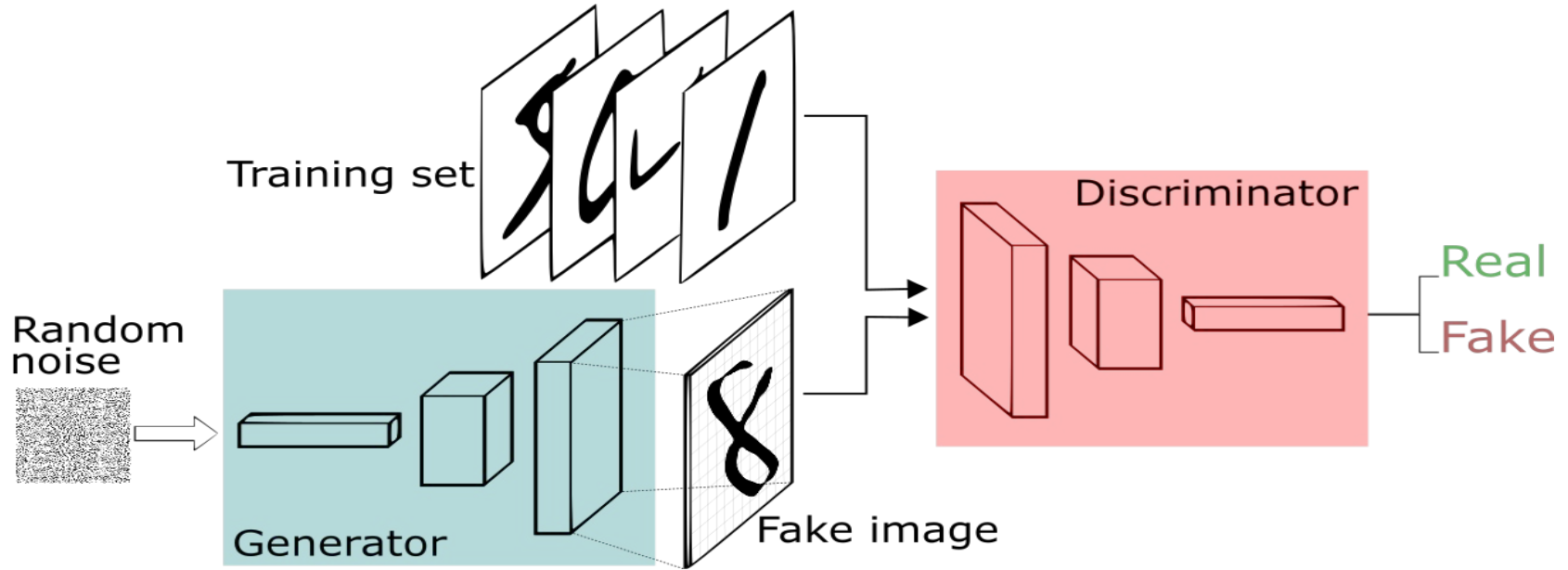
**Ian J. Goodfellow<sup>\*</sup>, Jean Pouget-Abadie<sup>†</sup>, Mehdi Mirza, Bing Xu, David Warde-Farley,  
Sherjil Ozair<sup>‡</sup>, Aaron Courville, Yoshua Bengio<sup>§</sup>**

Département d'informatique et de recherche opérationnelle  
Université de Montréal  
Montréal, QC H3C 3J7

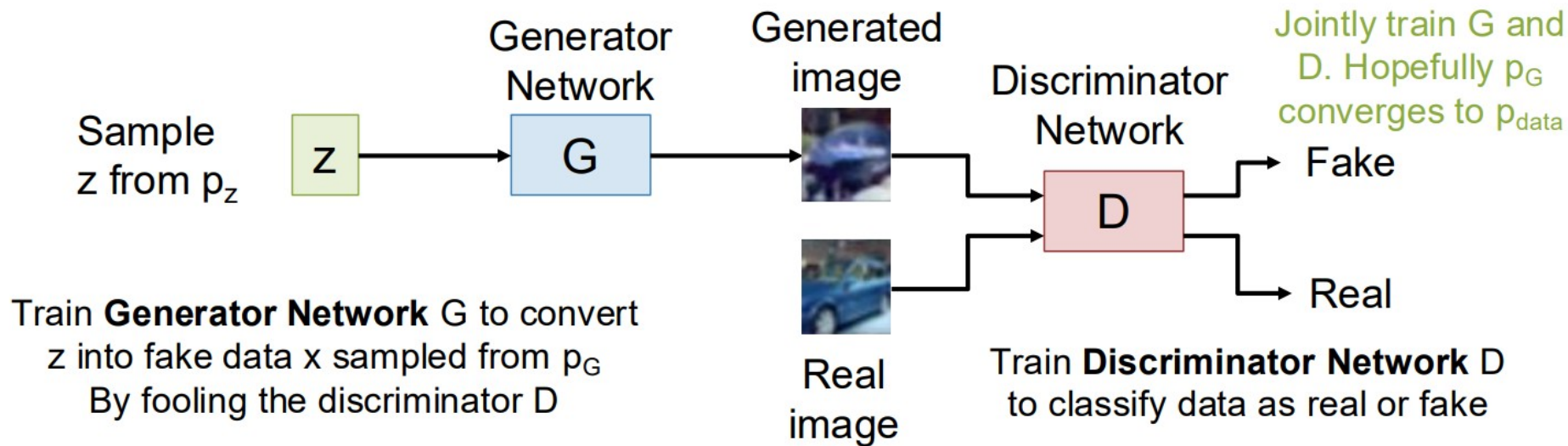
## Ian Goodfellow

Deep learner. Inventor of GANs. Lead author of  
<http://www.deeplearningbook.org>

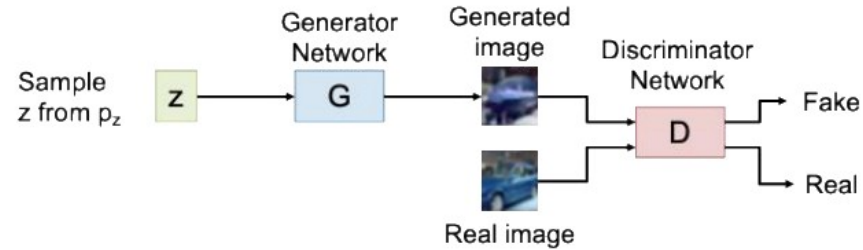
# Generative Adversarial Nets (GAN)



# Generative Adversarial Nets (GAN)



# Generative Adversarial Nets (GAN)



Jointly train generator G and discriminator D with a **minimax game**

$$\min_G \max_D \left( E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[ \log \left( 1 - D(G(z)) \right) \right] \right)$$

# Generative Adversarial Nets (GAN)

Jointly train generator  $G$  and discriminator  $D$  with a **minimax game**

Imagine fixing  $G$

Discriminator wants  $D(x) = 1$  for real data

Discriminator wants  $D(x) = 0$  for fake data

$$\min_G \max_D \left( E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[ \log \left( 1 - D(G(z)) \right) \right] \right)$$

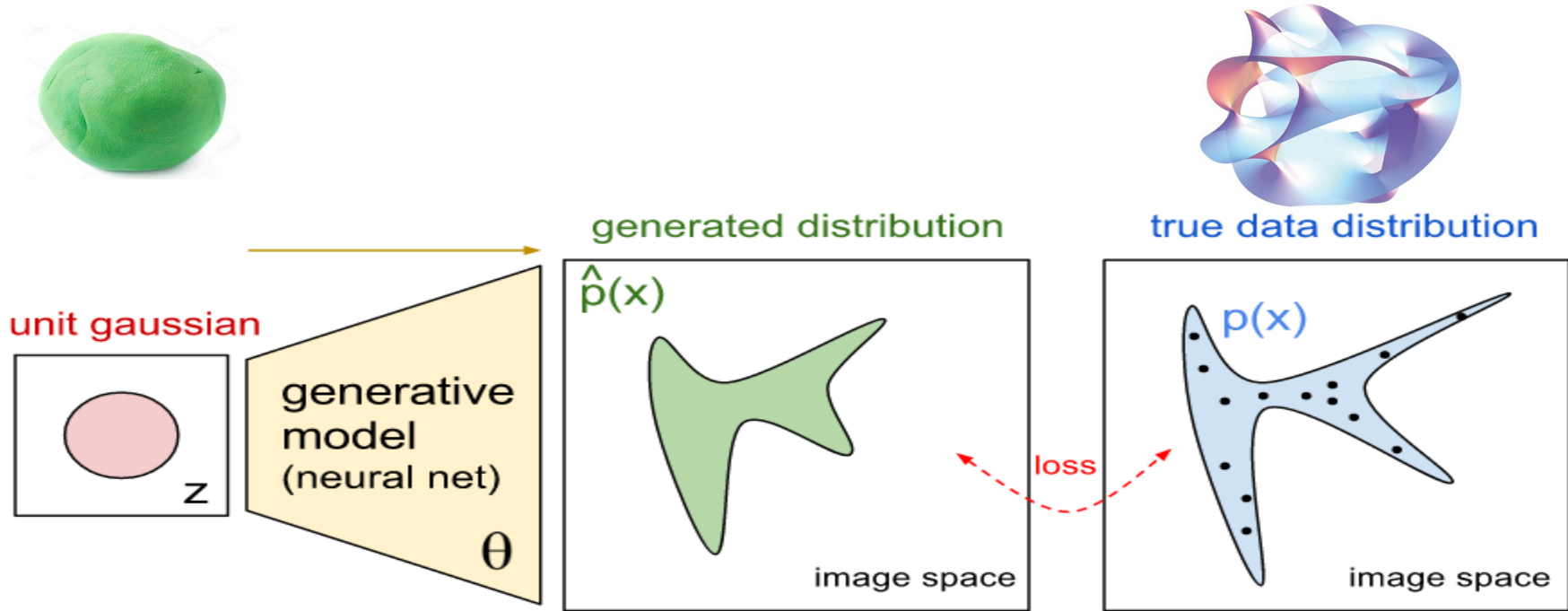
Imagine fixing  $D$

This term does not depend on  $G$

Generator wants  $D(x) = 1$  for fake data

$$\min_G \max_D \left( E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[ \log \left( 1 - D(G(z)) \right) \right] \right)$$

# Generative Adversarial Nets (GAN)

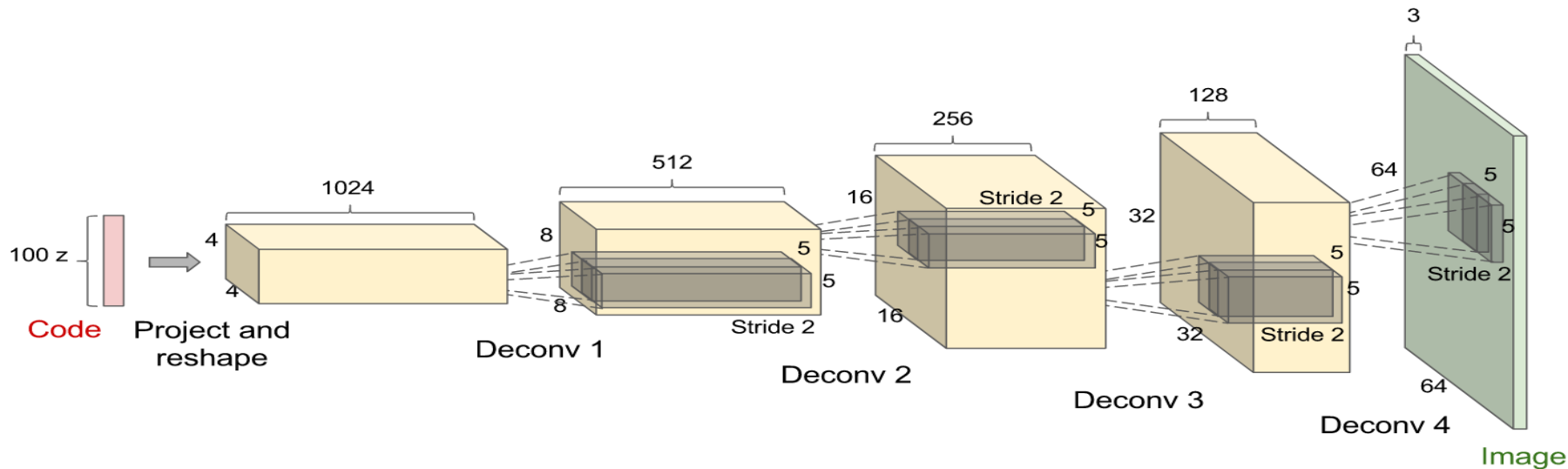


Kullback–Leibler divergence

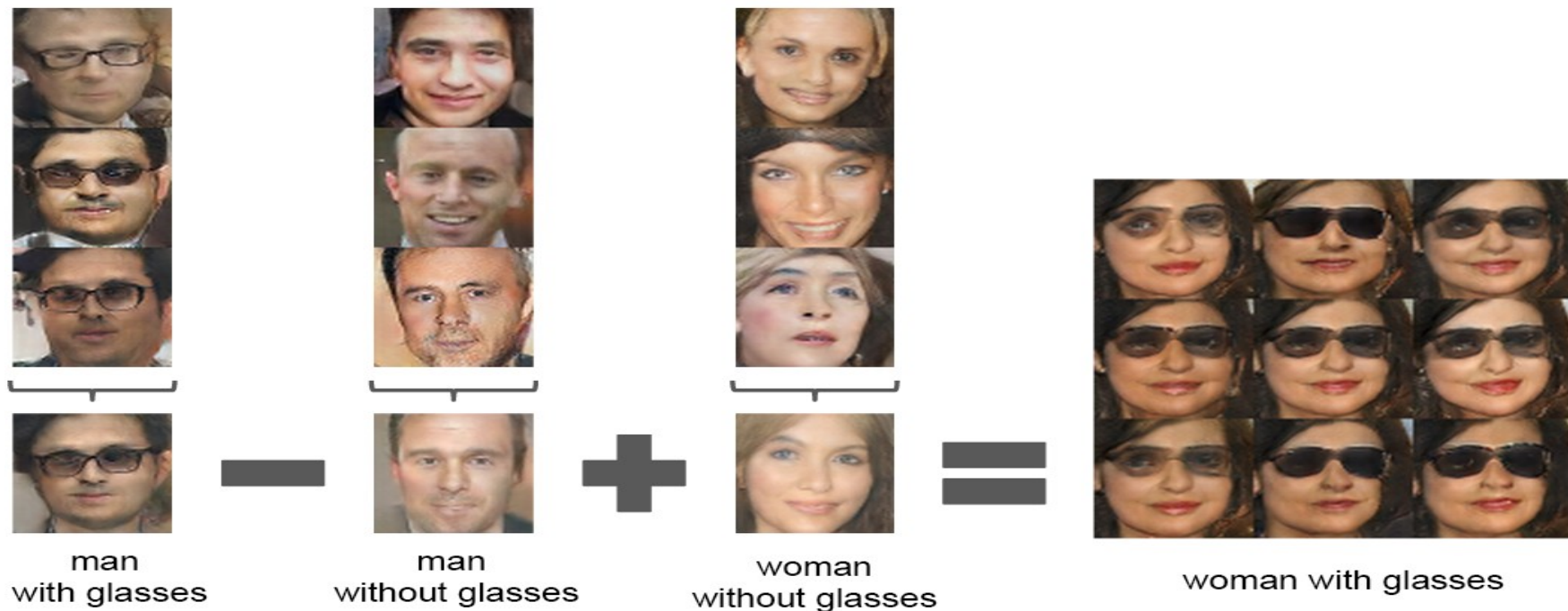
$$KL(\mathbb{P}_r \parallel \mathbb{P}_g) = \int \log \left( \frac{P_r(x)}{P_g(x)} \right) P_r(x) d\mu(x)$$

# Deep Convolutional GAN (dcgan, 2015)

Radford, A., Metz, L., & Chintala, S. (2015). Unsupervised representation learning with deep convolutional generative adversarial networks. *arXiv preprint arXiv:1511.06434*.



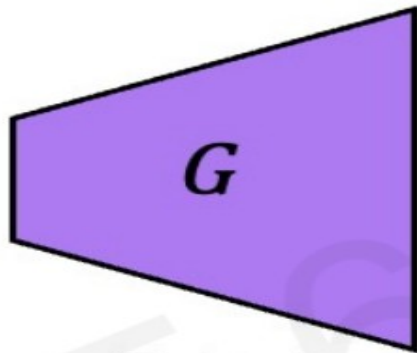
# Aritmética de caras



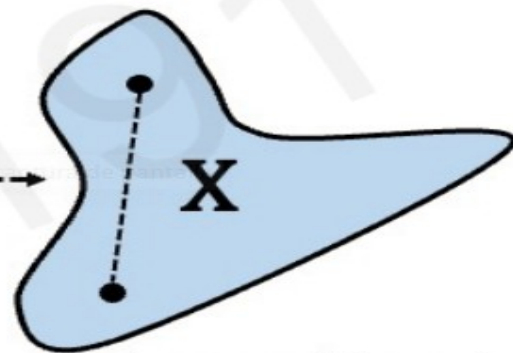
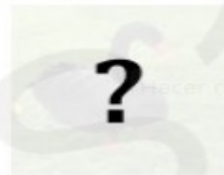
Radford, A., Metz, L., & Chintala, S. (2015). Unsupervised representation learning with deep convolutional generative adversarial networks. arXiv preprint arXiv:1511.06434.

# Interpolación

Gaussian noise  
 $z \sim \mathcal{N}(0,1)$



Trained  
generator

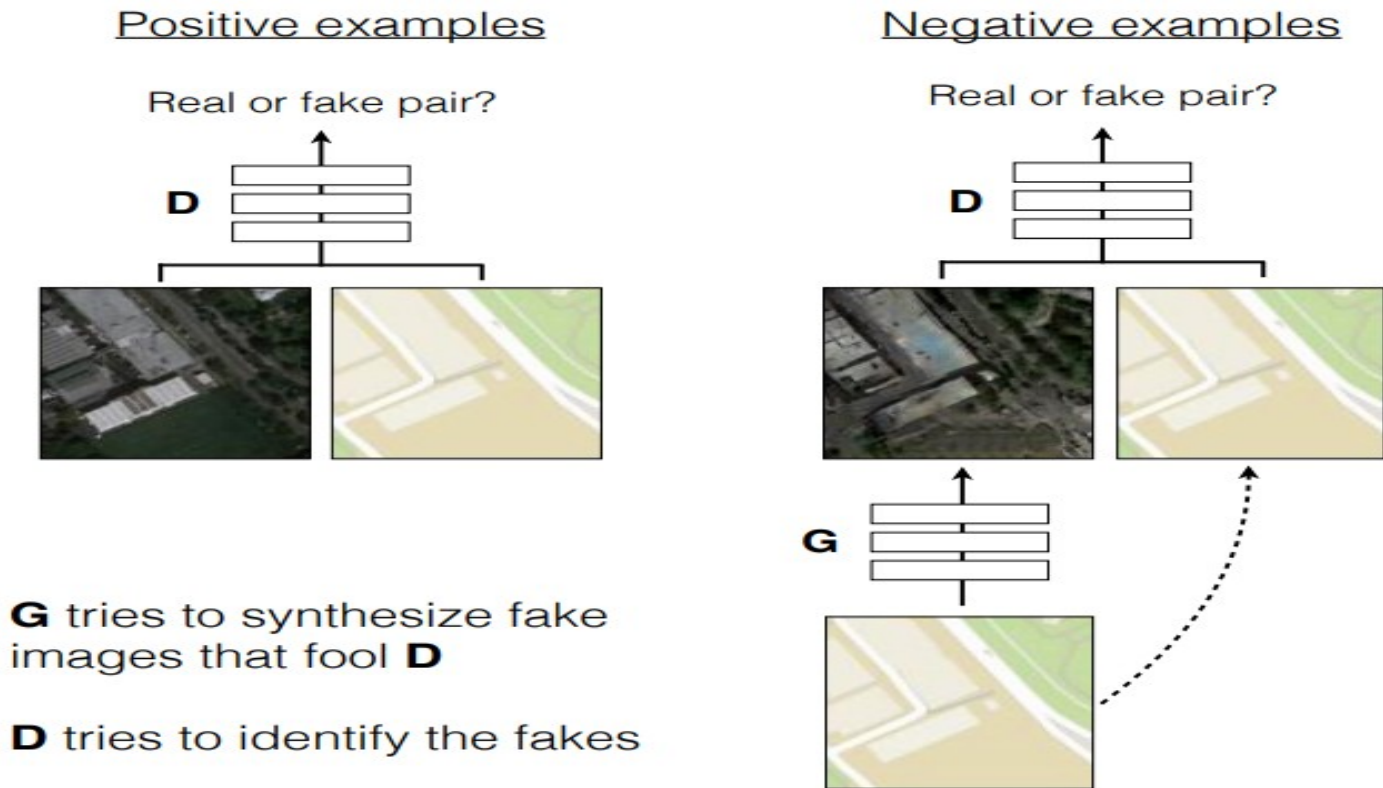


Learned target  
data distribution



# Algunas aplicaciones

# Image-to-Image Translation



# Image-to-Image Translation

Labels to Street Scene



input

output

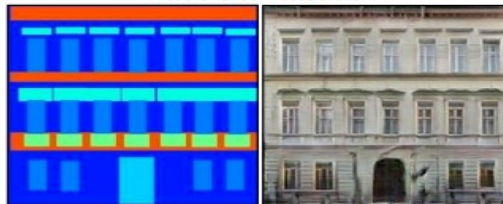
Aerial to Map



input

output

Labels to Facade



input

output

Day to Night



input

output

BW to Color



input

output

Edges to Photo



input

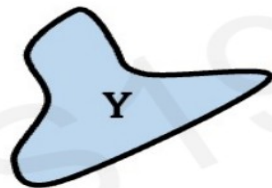
output

Isola, Phillip, et al. "Image-to-image translation with conditional adversarial networks." arXiv preprint arXiv:1611.07004 (2016).

# CycleGAN (2017)

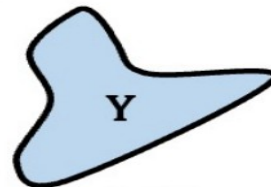
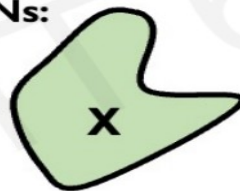
**GANs:**

Gaussian noise  
 $z \sim N(0,1)$



Gaussian noise  $\rightarrow$  target data manifold

**CycleGANs:**



data manifold  $X \rightarrow$  data manifold  $Y$



**Unpaired Image-to-Image Translation  
using Cycle-Consistent Adversarial Networks**

# CycleGAN

Cats  $\leftrightarrow$  Dogs



**itok\_msi** produced cats  $\leftrightarrow$  dogs CycleGAN results with a local+global discriminator and a smaller cycle loss.

Animal Transfiguration



**Tatsuya Hatanaka** trained our method to translate black bears to pandas. See more examples and download the models at the **website**. **Matt Powell** performed transfiguration between different species of birds



orange  $\rightarrow$  apple

# Turning Fortnite into PUBG with CycleGAN

CycleGAN conversion from Fortnite to PUBG



Fortnite

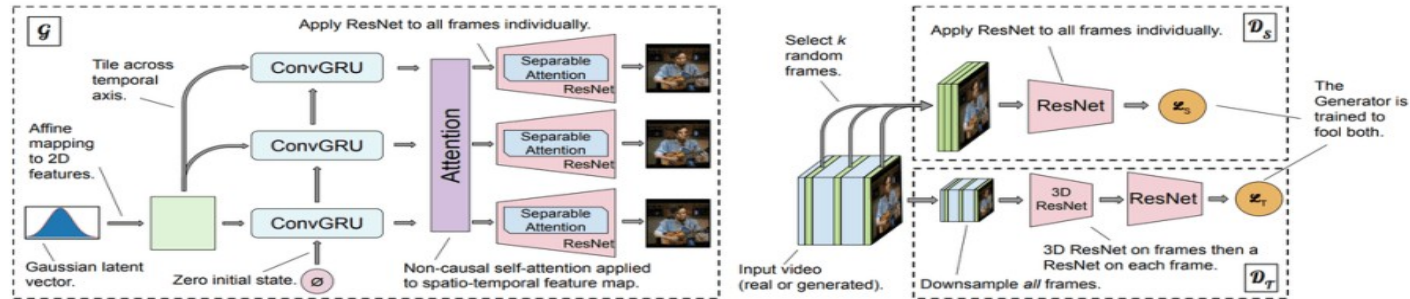


PUBG

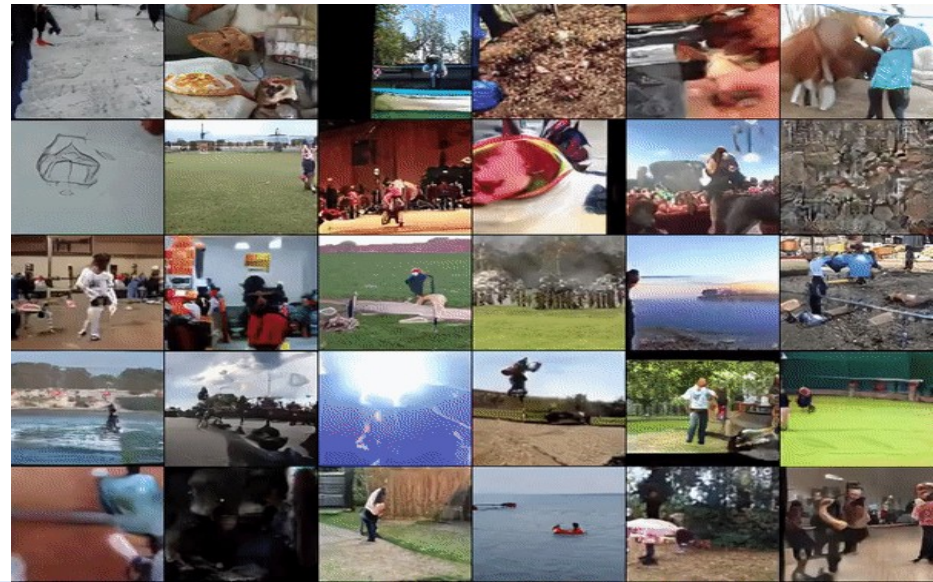


FortG? PUBnite?

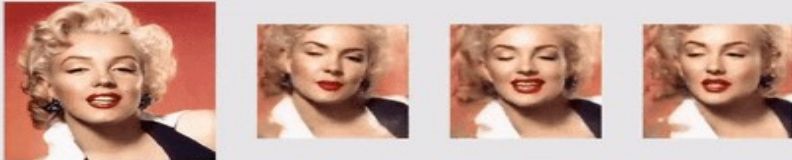
# Video: DVD-GAN (2019)



Big-GAN + Youtube +  
Discriminar en tiempo y espacio



# Living portraits



# Análisis Semántico en Rostros



Ejemplos de  
imágenes a  
clasificar



Caras  
generadas  
con GAN



Inteligencia Artificial

Iberoamerican Journal of Artificial Intelligence. Edited by Iberamia.

## Semantic analysis on faces using deep neural networks

Nicolás Federico Pellejero  
Guillermo Grinblat  
Lucas Uzal

DOI: <https://doi.org/10.4114/intartif.vol21iss61pp14-29>

**Keywords:** Deep, Learning, Emotion, Recognition.

# Exploiting GAN Internal Capacity for High-Quality Reconstruction of Natural Images

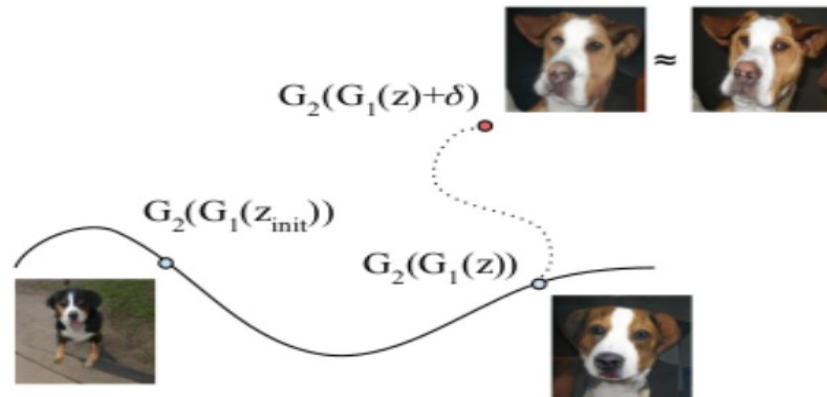
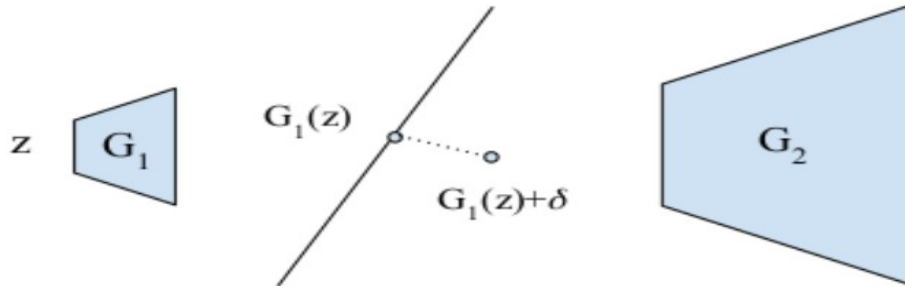
CONICET



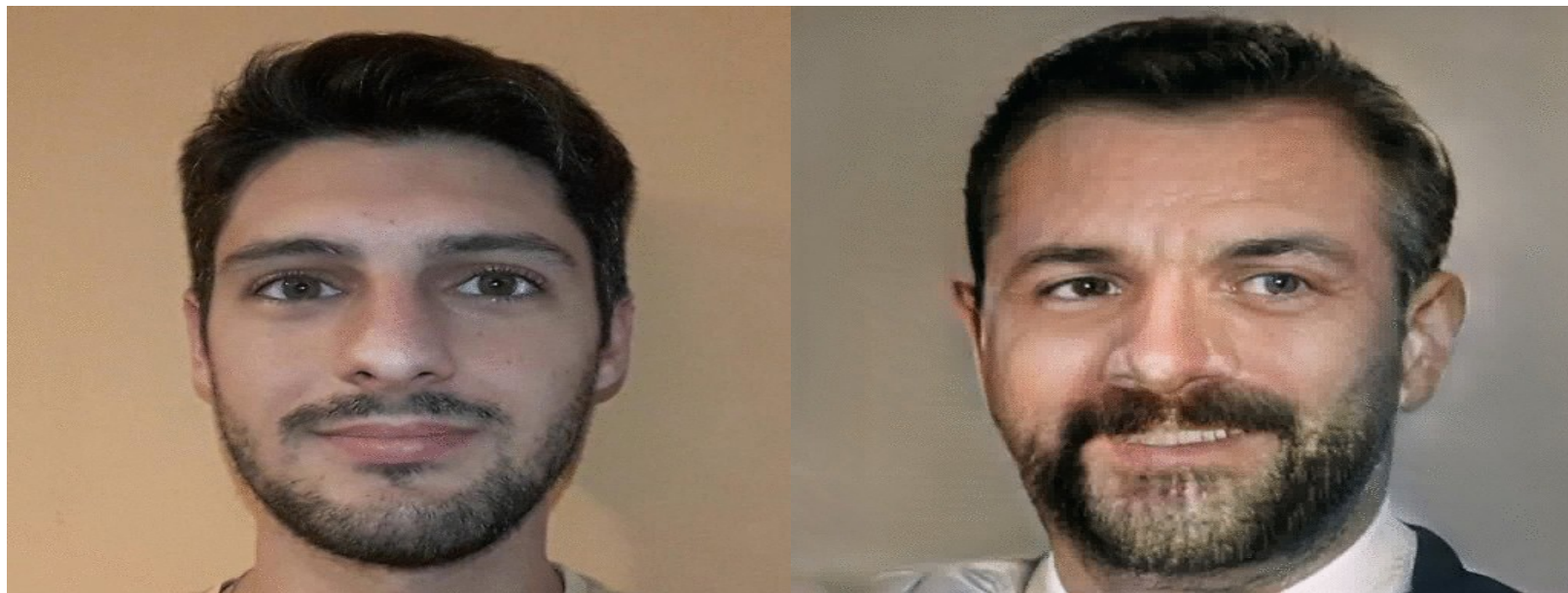
C I F A S I S



Marcos Pividori  
Guillermo  
Grinblat  
Lucas Uzal



# Generación de rostros y manipulación de expresiones



# Autoencoders y VAEs

# Autoencoders

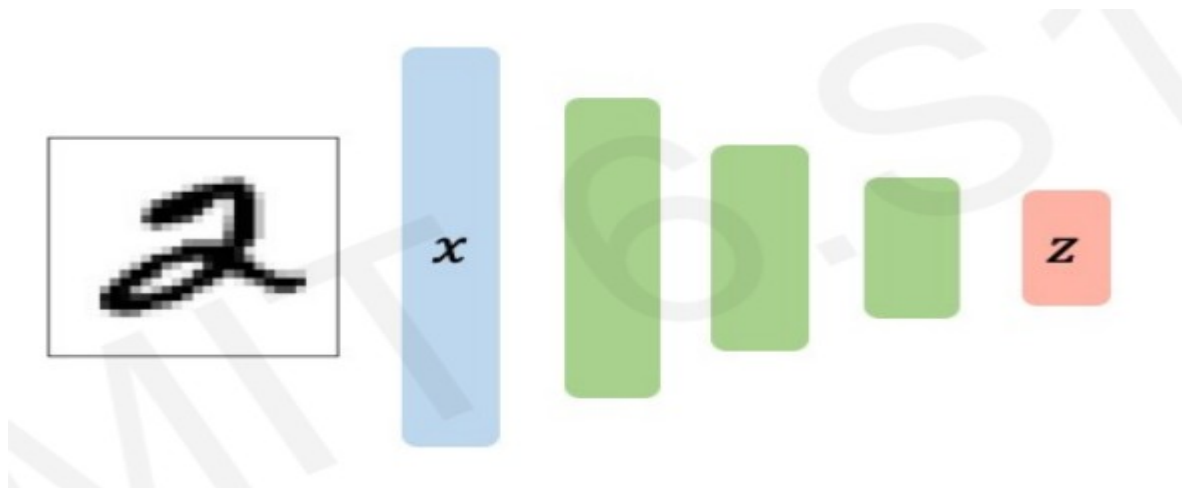
Variables latentes: Verdaderos factores de variación que explican un fenómeno observable.



*Myth of the Cave*

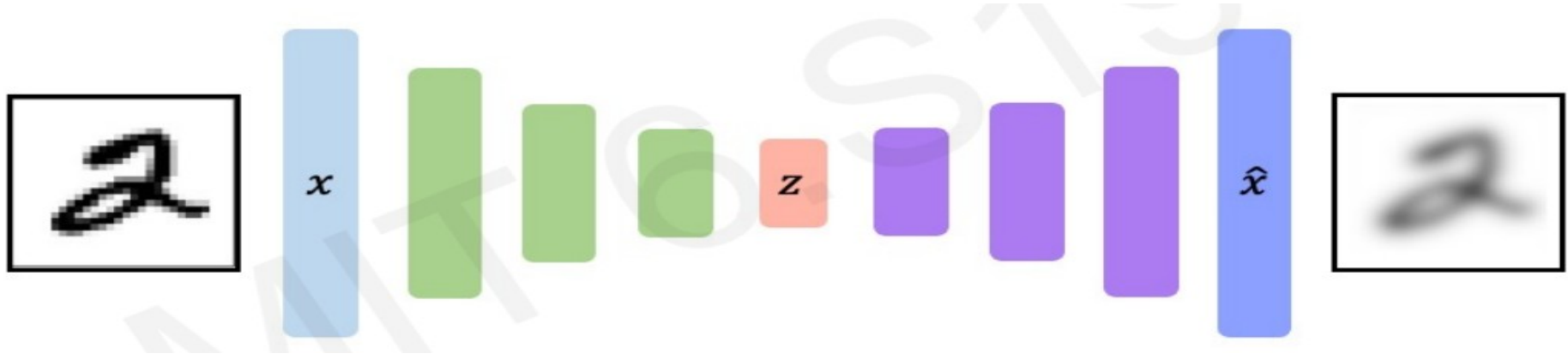
# Autoencoders

Encoder: modelo que transforma datos a vectores en un espacio de baja dimensión (espacio latente)



# Autoencoders

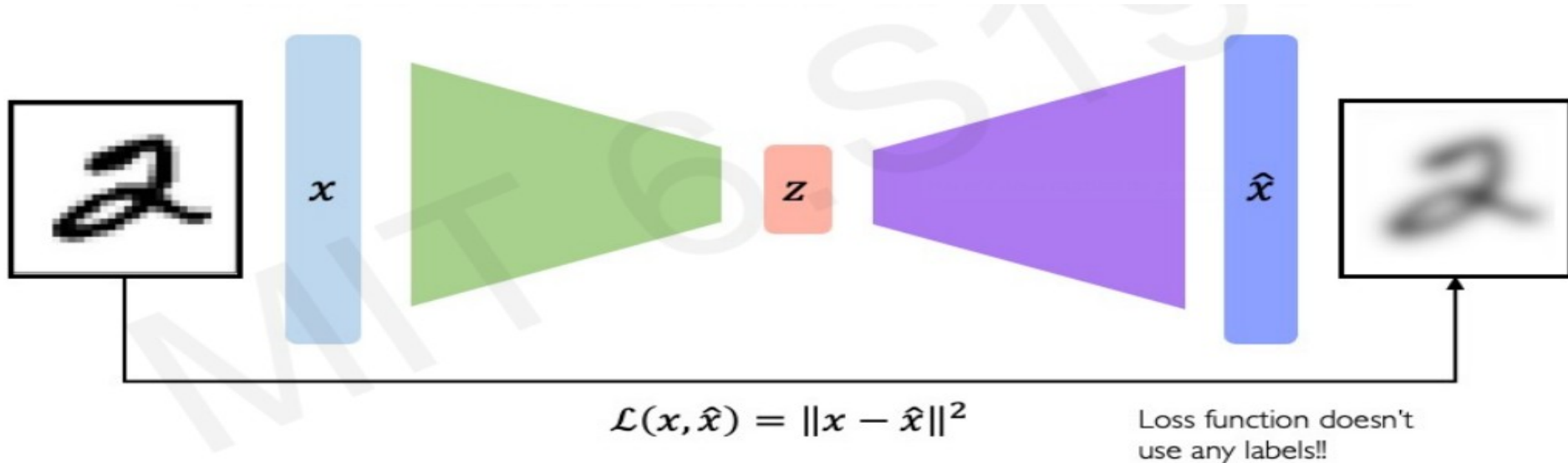
Decoder: modelo que transforma datos del espacio latente al espacio original



Autoencoder: modelo que transforma datos en sí mismos, pasando por un espacio latente.

# Autoencoders

Se utiliza una función costo que mide la calidad de la reproducción. No se necesitan datos con etiquetas.

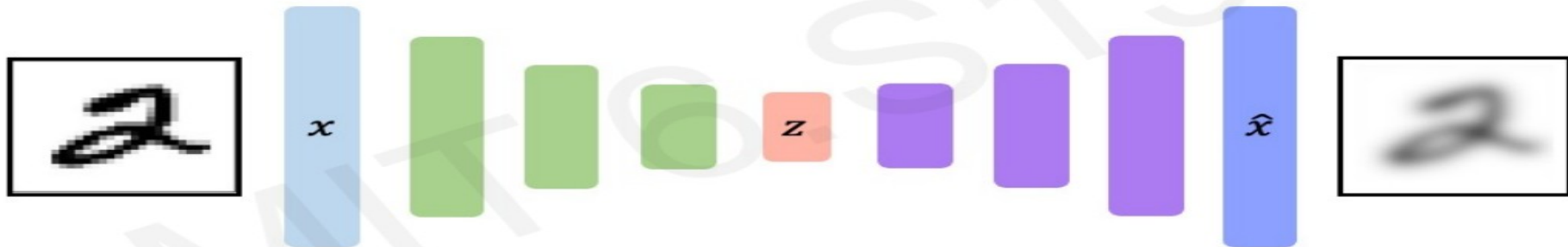


# Autoencoders: puntos clave

El cuello de botella en la red obliga a aprender una representación de baja dimensionalidad.

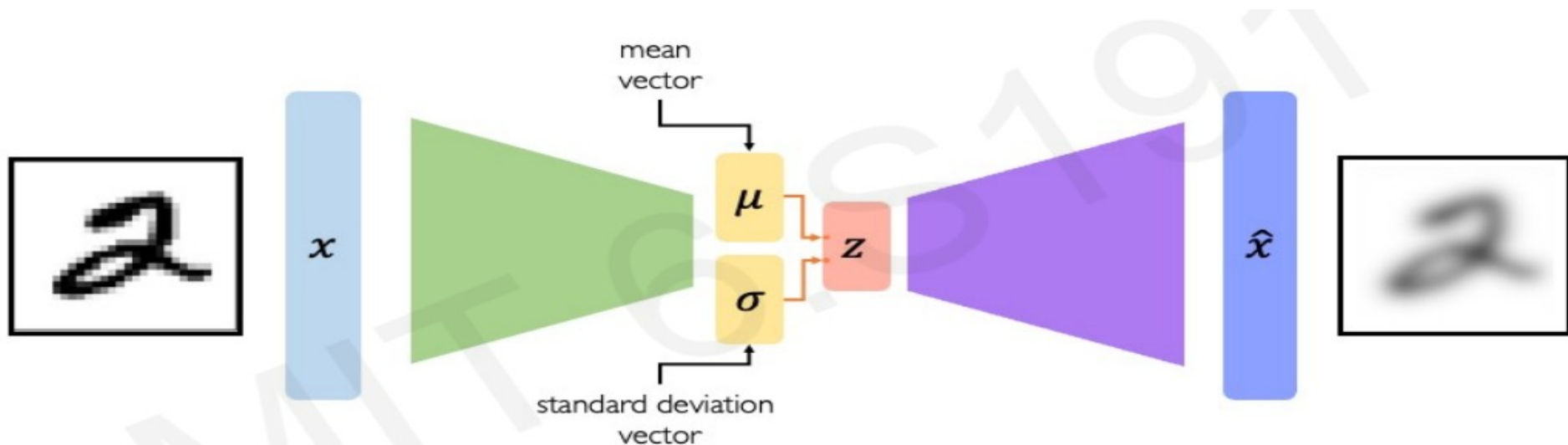
La función costo obliga a reproducir la mayor cantidad posible de información de la entrada

Resultado: proyecciones simples con gran información



# Autoencoders variacionales VAE

Autoencoder con mejora: asume que la representación en el espacio latente es sampleada de una distribución.



# Autoencoder: generación

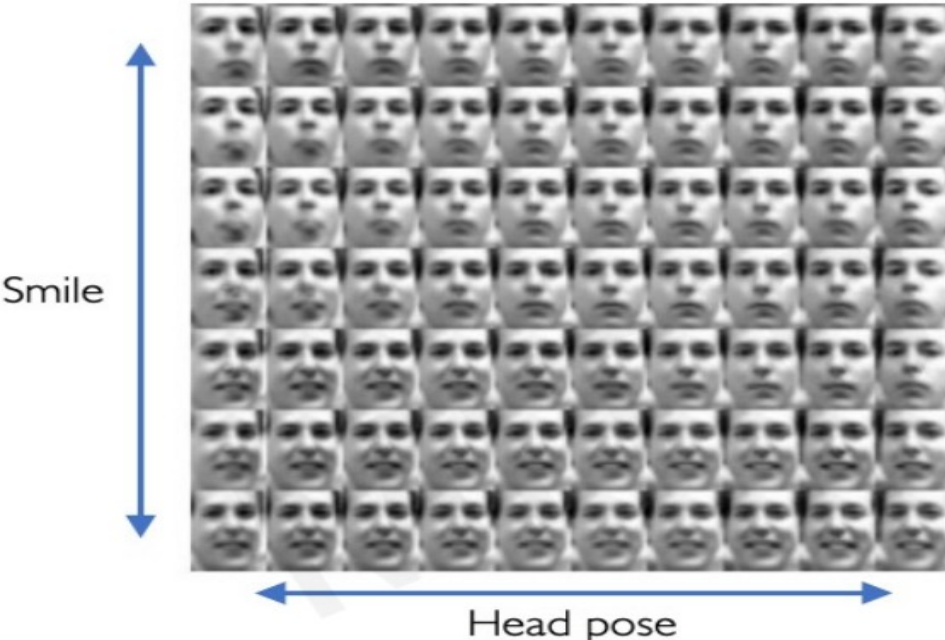
Se varía lentamente una de las variables latentes con las otras fijas.

Diferentes variables en  $Z$  suelen codificar diferentes factores de variación de los datos.



Head pose

# Autoencoder: generación



Queremos variables latentes independientes una de otra.

Se puede forzar al modelo para que aprenda variables “desenredadas”